Linear Search - Time and Space Complexity

Topic: TimeComplexity File: p1.pdf

PROBLEM

Given an unsorted array A of n values and a target x, return the index of x or -1 if it is absent.

IDEA

Inspect the elements from left to right. Stop immediately when the target is found.

PSEUDOCODE

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for i = 0 to n - 1:
    if A[i] == x: return i
return -1
```

EXAMPLE

A = [9, 4, 7, 2], x = 7. The checks are 9, then 4, then 7; return index 2 using zero-based indexing.

COMPLEXITY

Best time: O(1), when the first element matches. Average and worst time: O(n). Extra space: O(1).

TAKEAWAY

A single scan is linear because the number of comparisons grows in proportion to n.